

ZENITH

Technical Datasheet

The Kalmix ZENITH is a full-band GNSS surveying antenna for receiver systems that provide a compatible active-antenna input and antenna bias supply. This document consolidates the published RF, LNA, installation, and environmental specifications.

Antenna Characteristics

RF and phase-center specifications for multi-frequency GNSS reception.

Parameter	Specification
Frequency range	BeiDou: B1, B2, B3 GPS: L1, L2, L5 GLONASS: G1, G2, G3 Galileo: E1, E5a, E5b, E6 L-Band
Peak gain	5.5 dBi
Polarization	RHCP (Right-Hand Circular Polarization)
Axis ratio	≤ 3 dB
Phase center error	± 2 mm
Azimuth coverage	0 deg to 360 deg
VSWR (passive)	≤ 1.5
Impedance	50 ohm

LNA Characteristics

Published low-noise-amplifier parameters.

Parameter	Specification
LNA gain	L1: 38 $\pm$ 2 dB; L2: 40 $\pm$ 2 dB
Noise figure	≤ 1.8 dB
Output VSWR	≤ 2.0

Parameter	Specification
Passband ripple	+/- 2 dB
Operating voltage	DC 3.3 V to 12 V
Operating current	<= 50 mA

RF Interface and Installation

Parameter	Specification
RF connector	TNC-K female
Mounting thread	5/8 in x 11 TPI survey-standard thread
Receiver compatibility	GNSS receiver with an external active-antenna input and compatible antenna bias supply.
RF cable	Not included. Select separately for the receiver-side RF interface and installation path.

Environmental Characteristics

Parameter	Specification
Ingress protection	IP67-rated enclosure
Operating temperature	-40 C to +85 C
Storage temperature	-55 C to +85 C
Storage humidity	95% non-condensing

Installation note: verify antenna-bias voltage/current capability, RF connector compatibility, cable loss, grounding, and the required mounting hardware before deployment.